EXPLORATORY DATA ANALYSIS

Supermarket Sales Dataset

Student Name	Maliha
Internship Domain	Data Science
Organization	Devixo Solutions
Task Number	Task 01 - Exploratory Data Analysis (EDA)
GitHub Repository	[paste your repository link here]

Contents

1. Objective

2. Problem Statement

3. Dataset Description and Source

4. Technologies Used

5. Implementation Steps

6. Data Cleaning Process

7. Feature Analysis

8. Correlation Analysis

9. Visualizations

10. Business Insights

11. Business Recommendations

12. Code and Output Evidence

13. Challenges Faced

14. Conclusion

15. GitHub Repository Link

1. Objective

The objective of this task is to carry out a complete Exploratory Data Analysis (EDA) on a real-world retail dataset using Python. I wanted to clean the data, understand each feature, find relationships between variables, visualise the main patterns, and convert the results into business insights and practical recommendations.

2. Problem Statement

A supermarket chain with three branches wants to know how its sales are distributed and what could be improved. The questions I tried to answer are: Which branches, product lines, customers and time periods bring the most revenue? What drives the size of a customer's basket? What affects customer satisfaction? Are there weak areas, and what should the business do about them?

3. Dataset Description and Source

I used the **Supermarket Sales** dataset from Kaggle. It contains **1,000 transactions** and **17 columns** from three branches (A - Yangon, B - Mandalay, C - Naypyitaw) between 1 January and 30 March 2019. Each row is one invoice. Main columns are Branch, City, Customer type (Member / Normal), Gender, Product line (6 categories), Unit price, Quantity, Total, Date, Time, Payment (Cash / Ewallet / Credit card) and customer Rating (4 to 10). The dataset meets the task requirements: more than 1,000 records needed, with numerical and categorical features and enough information for a meaningful analysis. Source: Kaggle Supermarket Sales dataset (copy used: raw.githubusercontent.com/selva86/datasets/master/supermarket_sales.csv).

4. Technologies Used

Python 3, Pandas (data handling), NumPy (calculations), Matplotlib and Seaborn (visualisation), SciPy (statistical tests: ANOVA, Mann-Whitney U, chi-square) and Jupyter Notebook.

5. Implementation Steps

- **Import and understand:** loaded the CSV with Pandas, checked head, tail, shape, column names, data types and summary statistics.

- **Clean:** checked missing values and duplicates, converted Date to datetime, created Hour, Month, Weekday and Week features, checked category labels, and verified the formulas between the money columns.

- **Analyse features:** calculated min, max, mean, median and skew, category shares, and outliers using the IQR rule.

- **Correlation:** built a correlation matrix and heatmap on the suitable numerical features.

- **Visualise:** created 16 graphs, each one answering a specific question.

- **Test and interpret:** ran significance tests so that I only reported differences that could be real, then wrote 16 insights and 7 recommendations.

6. Data Cleaning Process

I cleaned the data before analysing it, because wrong data gives wrong conclusions. Here are my decisions:

- **Missing values:** there were none (0 across all 17 columns), so nothing was filled or dropped. Imputing values that do not exist would only add noise.

- **Duplicates:** 0 duplicate rows and all Invoice IDs are unique, so every row is a separate transaction.

- **Data types:** Date and Time were read as text. I converted Date to datetime and extracted Hour, Month, Weekday and Week for time analysis.

- **Inconsistent values:** category labels were consistent, with no spelling or spacing problems. Each Branch matches exactly one City.

- **Invalid values:** no negative numbers; Unit price is 10.08 to 99.96, Quantity 1 to 10, Rating 4 to 10, all reasonable.

- **Formula check (important):** cogs = unit price x quantity, Tax = 5% of cogs, Total = cogs + tax, gross income is identical to the Tax column, and gross margin percentage is the constant 4.7619% in every row. Because of this I used Total as my revenue measure and excluded these columns from correlation analysis.

Evidence - category labels check

```
# Inconsistent values: look at the labels of every categorical column
for col in ['Branch', 'City', 'Customer type', 'Gender', 'Product line', 'Payment']:
    print(col, '->', sorted(df[col].unique()))
print()
# Are there stray spaces or case problems?
for col in ['Branch', 'City', 'Customer type', 'Gender', 'Product line', 'Payment']:
    assert (df[col] == df[col].str.strip()).all()
print('No leading/trailing space problems.')
print('Branch/City mapping:'); print(df.groupby('Branch')['City'].unique())
```

```
Branch -> ['A', 'B', 'C']
City -> ['Mandalay', 'Naypyitaw', 'Yangon']
Customer type -> ['Member', 'Normal']
Gender -> ['Female', 'Male']
Product line -> ['Electronic accessories', 'Fashion accessories', 'Food and beverages', 'Health and beauty', 'Home and
Payment -> ['Cash', 'Credit card', 'Ewallet']

No leading/trailing space problems.
Branch/City mapping:
Branch
A       [Yangon]
B     [Mandalay]
C    [Naypyitaw]
Name: City, dtype: object
```

Evidence - formula and validity checks

```
# Invalid values: do the money columns follow the dataset's own rules?
print('Unit price range :', df['Unit price'].min(), '-', df['Unit price'].max())
print('Quantity range   :', df['Quantity'].min(), '-', df['Quantity'].max())
print('Rating range     :', df['Rating'].min(), '-', df['Rating'].max())
print('Any negative values:', (df[['Unit price','Quantity','Total','cogs']] < 0).any().any())

print('\ncogs == unit price x quantity      :', np.allclose(df['cogs'], df['Unit price']*df['Quantity']))
print('Tax 5% == 5% of cogs               :', np.allclose(df['Tax 5%'], 0.05*df['cogs']))
print('Total == cogs + tax                :', np.allclose(df['Total'], df['cogs'] + df['Tax 5%']))
print('gross income == Tax 5%             :', np.allclose(df['gross income'], df['Tax 5%']))
print('gross margin % unique values       :', df['gross margin percentage'].unique())
```

```
Unit price range : 10.08 - 99.96
Quantity range   : 1 - 10
Rating range     : 4.0 - 10.0
Any negative values: False

cogs == unit price x quantity      : True
Tax 5% == 5% of cogs               : True
Total == cogs + tax                : True
gross income == Tax 5%             : True
gross margin % unique values       : [4.76190476]
```

7. Feature Analysis

The table below shows the main numerical features. **Total** has a mean ($322.97) well above its median ($253.85) and a skew of about 0.89, which means a minority of large baskets lifts the average. Unit price, Quantity and Rating are symmetric. Only Total has outliers: 9 baskets above $991.74, all of them 10 items at roughly $95-99 each, so they are genuine purchases and I kept them.

Evidence - numerical feature summary

```
num_features = ['Unit price', 'Quantity', 'Total', 'Rating']
summary = df[num_features].agg(['min', 'max', 'mean', 'median', 'std', 'skew']).T.round(2)
summary
```

```
              min      max    mean  median     std  skew
Unit price  10.08    99.96   55.67   55.23   26.49  0.01
Quantity     1.00    10.00    5.51    5.00    2.92  0.01
Total       10.68  1042.65  322.97  253.85  245.89  0.89
Rating       4.00    10.00    6.97    7.00    1.72  0.01
```

Evidence - outlier detection (IQR rule)

```
# Outliers with the IQR rule
rows = []
for col in num_features:
    q1, q3 = df[col].quantile([0.25, 0.75]); iqr = q3 - q1
    low, high = q1 - 1.5*iqr, q3 + 1.5*iqr
    n = ((df[col] < low) | (df[col] > high)).sum()
    rows.append([col, round(low,2), round(high,2), n])
pd.DataFrame(rows, columns=['Feature', 'Lower fence', 'Upper fence', 'Outliers'])
```

```
      Feature  Lower fence  Upper fence  Outliers
0  Unit price       -34.72       145.53         0
1    Quantity        -4.50        15.50         0
2       Total      -395.97       991.74         9
3      Rating         1.00        13.00         0
```

Evidence - category counts and shares

```
# Categorical features: counts and share
for col in ['Branch', 'Customer type', 'Gender', 'Product line', 'Payment']:
    t = df[col].value_counts()
    print(pd.DataFrame({'count': t, 'share %': (t/len(df)*100).round(1)}))
    print()
```

```
        count  share %
Branch
A         340     34.0
B         332     33.2
C         328     32.8

               count  share %
Customer type
Member           501     50.1
Normal           499     49.9

        count  share %
Gender
Female    501     50.1
Male      499     49.9

                        count  share %
Product line
Fashion accessories       178     17.8
Food and beverages        174     17.4
Electronic accessories    170     17.0
Sports and travel         166     16.6
Home and lifestyle        160     16.0
Health and beauty         152     15.2

             count  share %
Payment
Ewallet        345     34.5
Cash           344     34.4
Credit card    311     31.1
```

8. Correlation Analysis

I left out the constant margin column and the formula-based columns (cogs, tax, gross income), because they would only show trivial perfect correlations.

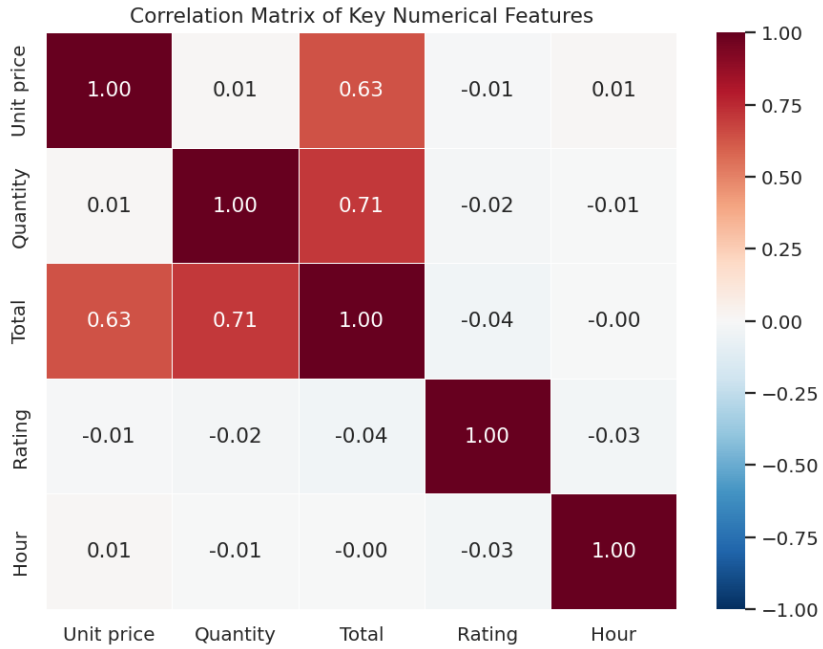


Figure 1 - Correlation matrix of key numerical features

- **Strong positive:** Quantity and Total (0.71), Unit price and Total (0.63). Basket value is price multiplied by quantity.

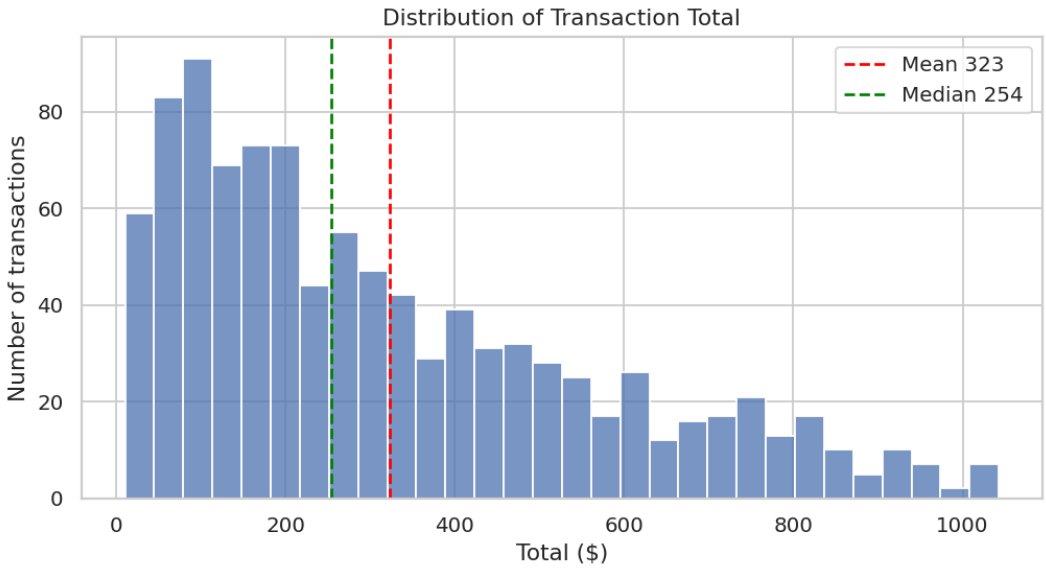
- **Strong negative:** none. The lowest value is Total vs Rating at -0.04, which is essentially zero.

- **Weak relationships:** Rating shows no relationship with price, quantity, total or hour (all below 0.04 in size). Unit price and Quantity are also unrelated (0.01).

- **Most important relationship:** price and quantity independently drive basket size. Satisfaction is not explained by any recorded variable.

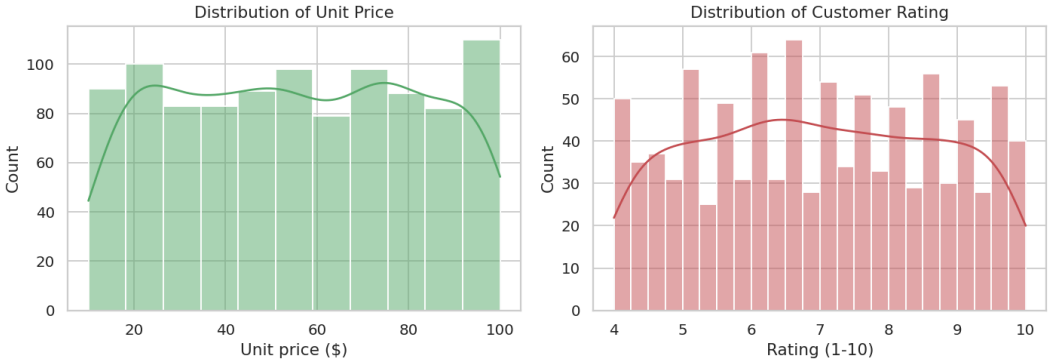
9. Visualizations

Figure 2 - Histogram - distribution of transaction total



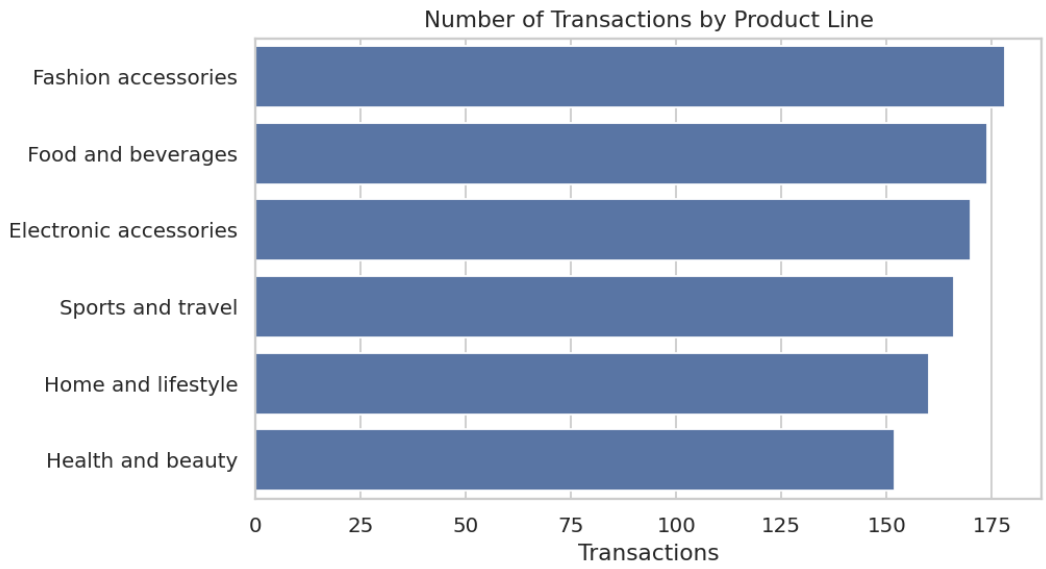
Most baskets are small or medium and the right tail reaches about $1,000. The mean is above the median because of this tail.

Figure 3 - Distribution plots - unit price and rating



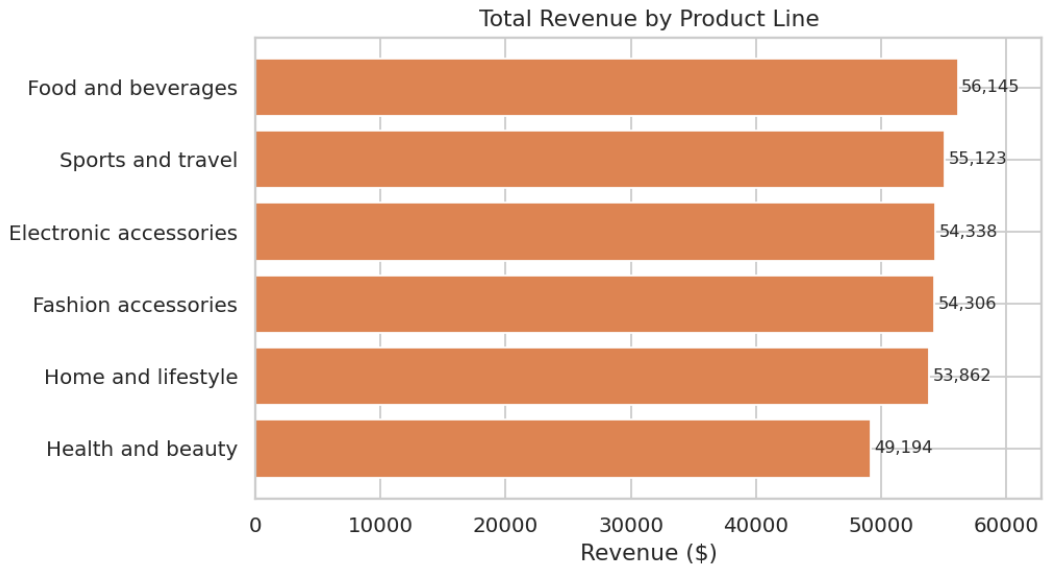
Prices spread evenly between $10 and $100 and ratings between 4 and 10. No rating is below 4.

Figure 4 - Count plot - transactions per product line



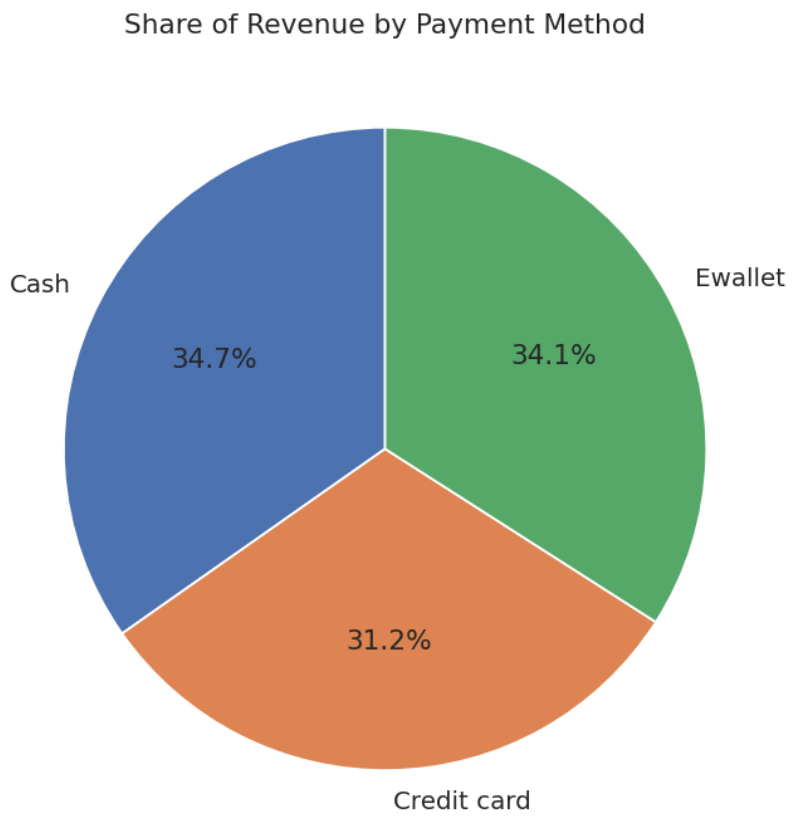
Fashion accessories and Food and beverages have the most transactions, Health and beauty the fewest (152).

Figure 5 - Bar chart - revenue by product line



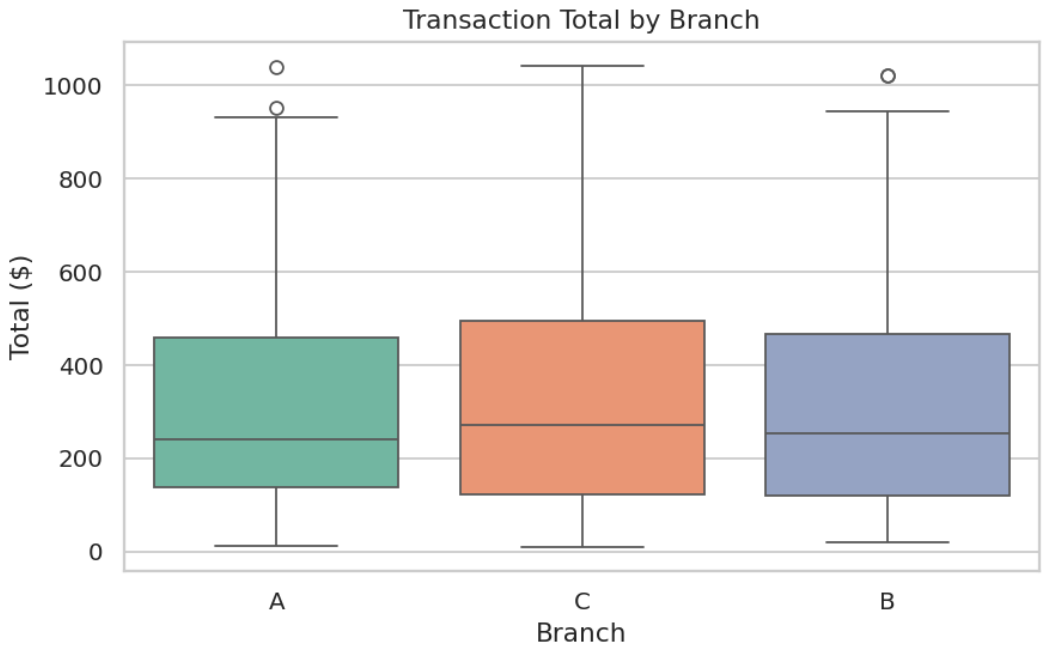
Food and beverages earns the most ($56.1k) and Health and beauty the least ($49.2k). The gap is due to transaction counts, not basket size.

Figure 6 - Pie chart - revenue share by payment method



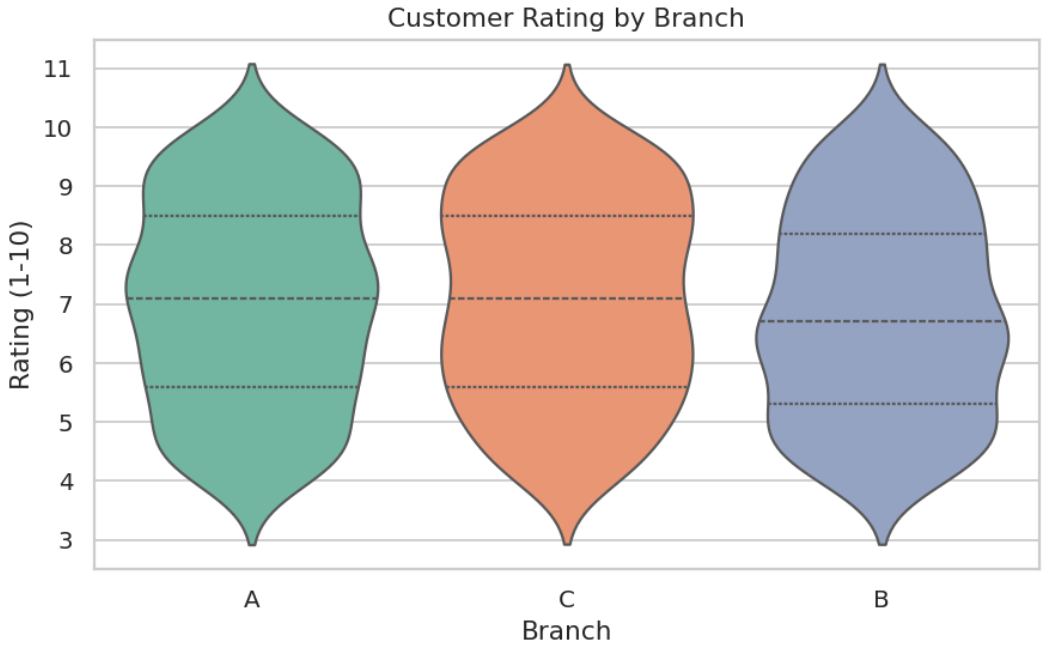
Cash 34.7%, Ewallet 34.1%, Credit card 31.2%. Payment use is well balanced.

Figure 7 - Box plot - transaction total by branch



The three branches have nearly the same median and spread. High-value outliers appear in every branch.

Figure 8 - Violin plot - rating by branch



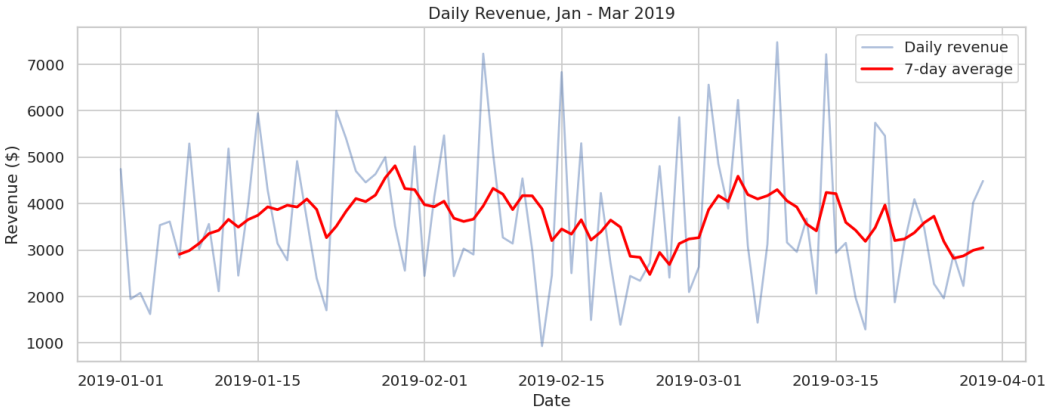
Branch B is centred slightly lower (average 6.82) than A (7.03) and C (7.07), but the difference is not statistically significant (p = 0.13).

Figure 9 - Scatter plot - unit price vs total, coloured by quantity



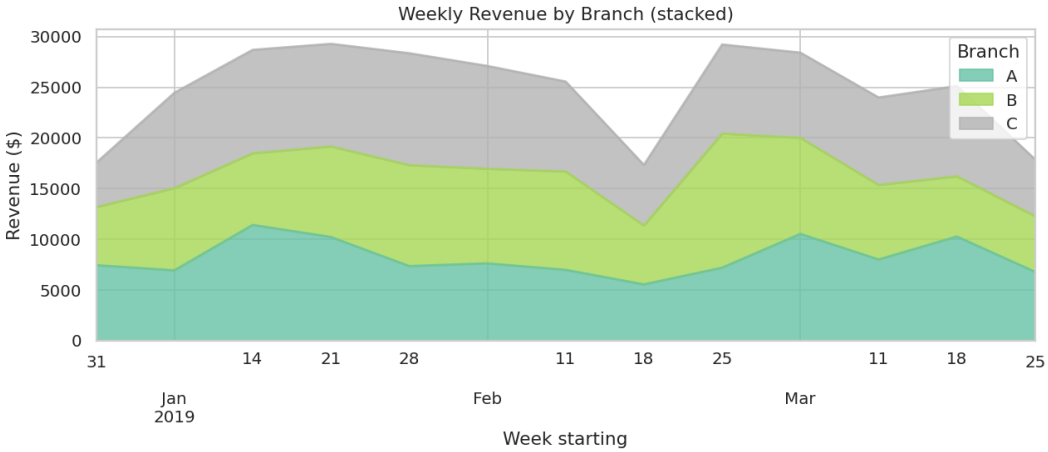
For the same unit price, a higher quantity gives a higher total. The fan shape is the price x quantity relationship.

Figure 10 - Line chart - daily revenue with 7-day average



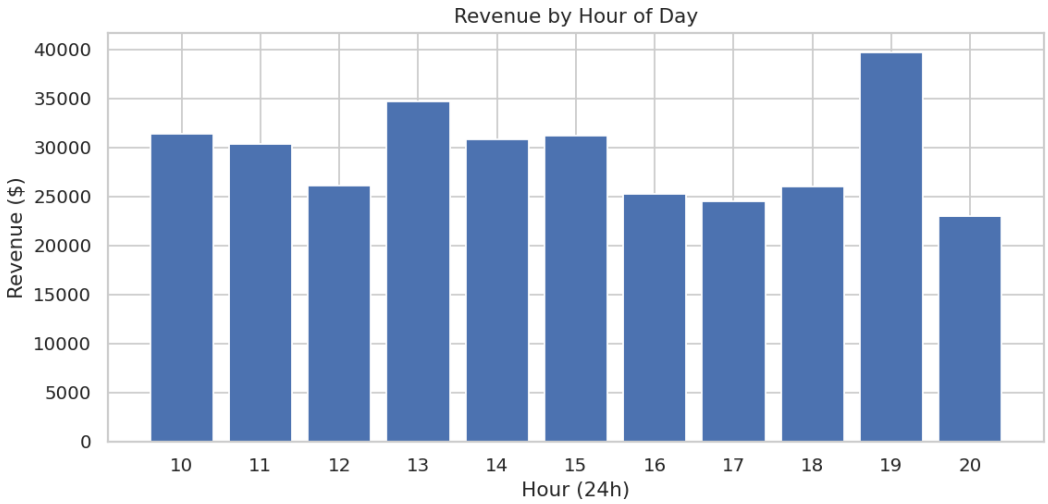
Daily sales are very uneven, and the 7-day average shows no clear growth through the quarter.

Figure 11 - Area chart - weekly revenue by branch



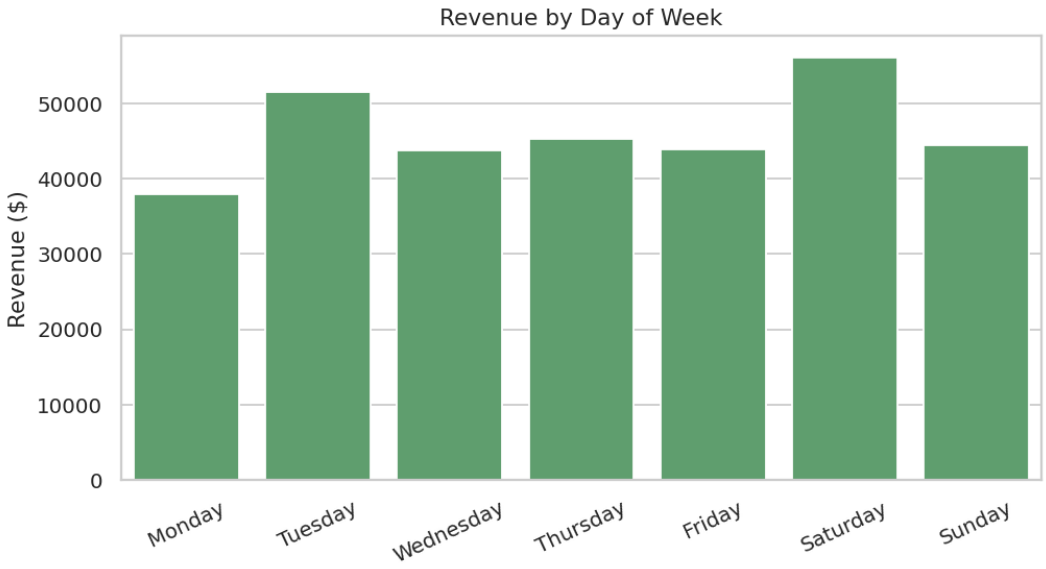
No branch leads in every week. The low first and last weeks are partial weeks.

Figure 12 - Bar chart - revenue by hour



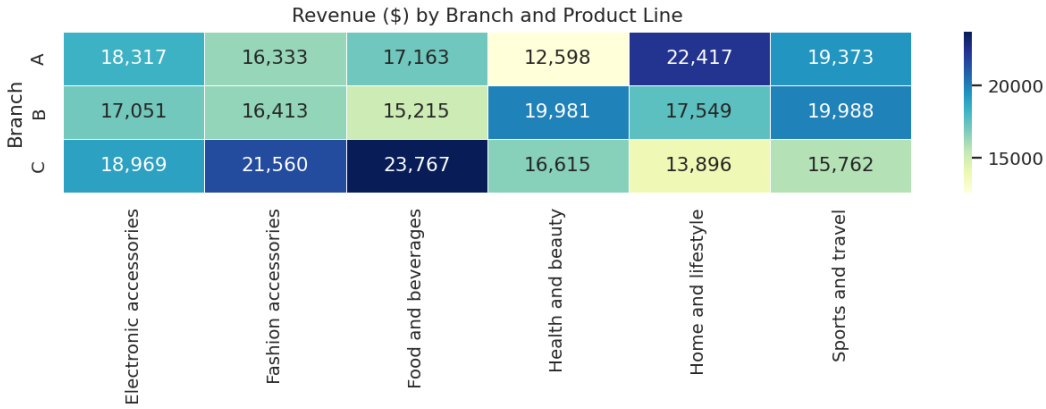
19:00 is the peak hour ($39.7k, 113 transactions). 16:00-17:00 and 20:00 are the quietest.

Figure 13 - Bar chart - revenue by weekday



Saturday is the best day ($56.1k) and Monday the weakest ($37.9k).

Figure 14 - Heatmap - revenue by branch and product line



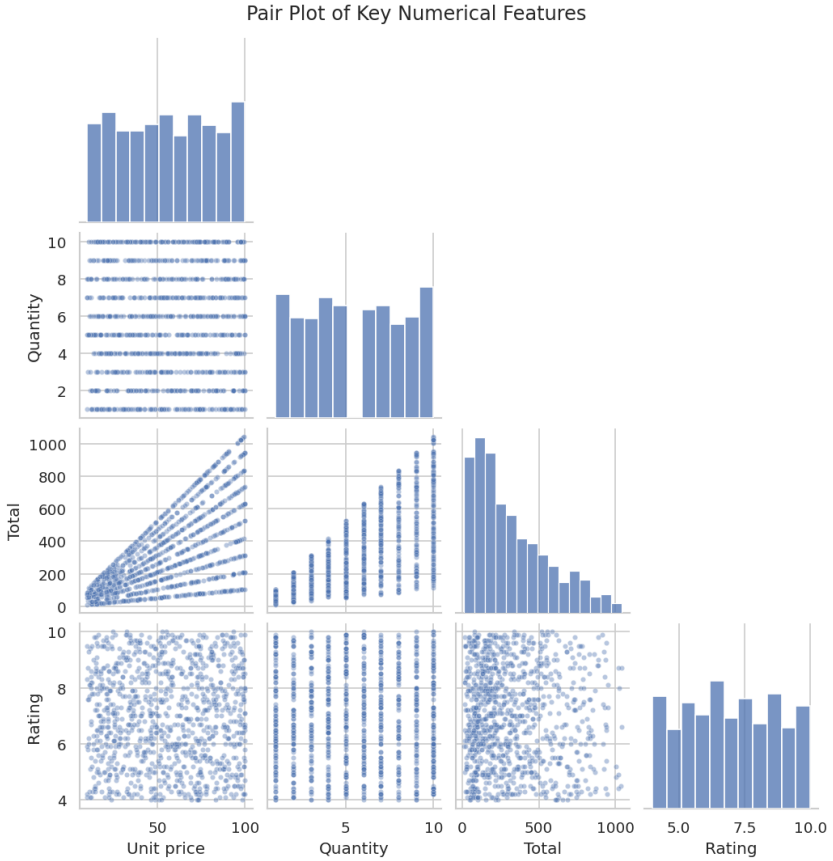
Branch B is strong in Health and beauty, C in Food and beverages and Fashion, A in Home and lifestyle. These patterns are not statistically proven (chi-square p = 0.32).

Figure 15 - Grouped bar - revenue by product line and gender



Women spend more on Food, Fashion and Home, men on Health and beauty, and Electronics is equal. The category mix difference is not statistically significant (p = 0.33).

Figure 16 - Pair plot - key numerical features



Confirms that Total is linked to price and quantity, while Rating shows no pattern against anything.

10. Business Insights

I first tested the main group differences, so that I report as real only what the data supports. The results are shown below (p-value below 0.05 means significant). None of the tested differences reached significance, which is itself an important finding: this business is very evenly balanced.

Evidence - significance tests

```
member = df[df['Customer type']=='Member']; normal = df[df['Customer type']=='Normal']
female = df[df['Gender']=='Female']; male = df[df['Gender']=='Male']

tests = {
 'Basket size: Member vs Normal (Mann-Whitney)': stats.mannwhitneyu(member['Total'], normal['Total']).pvalue,
 'Basket size: Female vs Male (Mann-Whitney)'  : stats.mannwhitneyu(female['Total'], male['Total']).pvalue,
 'Basket size across Branches (ANOVA)'         : stats.f_oneway(*[x['Total'] for _, x in df.groupby('Branch')]).pvalu
 'Rating across Branches (ANOVA)'              : stats.f_oneway(*[x['Rating'] for _, x in df.groupby('Branch')]).pval
 'Basket size across Product lines (ANOVA)'    : stats.f_oneway(*[x['Total'] for _, x in df.groupby('Product line')])
 'Rating across Product lines (ANOVA)'         : stats.f_oneway(*[x['Rating'] for _, x in df.groupby('Product line')]
}
ct = pd.crosstab(df['Product line'], df['Gender'])
tests['Product line mix differs by Gender (chi-square)'] = stats.chi2_contingency(ct)[1]
ct2 = pd.crosstab(df['Branch'], df['Product line'])
```

	p-value	significant
Basket size: Member vs Normal (Mann-Whitney)	0.591	False
Basket size: Female vs Male (Mann-Whitney)	0.101	False
Basket size across Branches (ANOVA)	0.413	False
Rating across Branches (ANOVA)	0.126	False
Basket size across Product lines (ANOVA)	0.890	False
Rating across Product lines (ANOVA)	0.746	False
Product line mix differs by Gender (chi-square)	0.332	False
Product line mix differs by Branch (chi-square)	0.316	False

1. **Revenue is evenly spread across branches.** Total revenue is about $322,967. Branch C leads with $110.6k (34.2%), then A with $106.2k and B with $106.2k. The ANOVA on basket size gives p = 0.41, so the branch gap is small and mostly comes from Branch C's slightly bigger average basket ($337 vs $312 and $320).

2. **Baskets are right-skewed.** The average basket is $323 but the median is only $254, so a minority of large purchases lifts the mean. The top 10% of transactions bring in about 26% of all revenue.

3. **Basket value is driven by quantity and price separately.** `Total` correlates with `Quantity` (0.71) and `Unit price` (0.63), while price and quantity are unrelated to each other (0.01). Customers do not cut quantity when prices are higher, so both are separate levers for growth.

4. **Customer rating is not explained by anything in the data.** Rating has |r| < 0.04 with price, quantity, total and hour. Satisfaction (average 6.97) does not depend on how much someone spends, so the drivers must be things not recorded here, like service or waiting time.

5. **About 1 in 6 customers is unhappy and 1 in 6 is delighted.** 15.3% of ratings are below 5 and 16.6% are 9 or above. Ratings never go below 4, so the scale is used in a narrow range.

6. **Branch B has the lowest satisfaction.** Its average rating is 6.82 vs 7.07 (C) and 7.03 (A). The test gives p = 0.13, so it is *not* statistically proven, but it is the one branch worth watching.

7. **Food and beverages is the top category, Health and beauty the weakest.** Food and beverages earns $56.1k, Health and beauty $49.2k. Basket sizes across product lines are statistically the same (p = 0.89), so the ranking is due to how many transactions each line gets (174 vs 152).

8. **Women and men differ in category spending, but the evidence is weak.** Women spend more on Food and beverages ($33.2k vs $23.0k), Fashion ($30.4k vs $23.9k) and Home and lifestyle ($30.0k vs $23.8k), while men spend more on Health and beauty ($30.6k vs $18.6k). Electronics is equal.

However, the chi-square test on category mix by gender gives p = 0.33, so with only 1,000 rows this could be chance. I treat it as a hypothesis worth testing, not a proven fact. Women also buy slightly more items per basket (5.73 vs 5.29), but their basket size gap is not significant either (p = 0.10).

9. **Branches show different category strengths, again not proven.** Branch B earns $20.0k in Health and beauty vs $12.6k in A. Branch C earns $23.8k in Food and beverages but only $13.9k in Home and lifestyle, while A earns $22.4k there. The chi-square test on category mix by branch gives p = 0.32, so these gaps may be random variation. They are good places to run small experiments, not to make big changes.

10. **The evening is the busiest time.** 19:00 is the best hour with 113 transactions and about $39.7k. The slowest hours are 16:00-17:00 and 20:00 (roughly $23-25k each, against $39.7k at 19:00). Staffing and stock should follow this curve.

11. **Saturday is the best day and Monday the worst.** Saturday brings $56.1k, Monday only $37.9k. Monday also has the fewest transactions (125) and the smallest basket quantity (5.10).

12. **Monthly sales were flat to slightly down, not growing.** January $116.3k, March $109.5k, February $97.2k. February is shorter, so I compared per trading day: roughly $3.7k (Jan), $3.5k (Feb), $3.6k (Mar). There is no growth trend in the quarter.

13. **The membership card does not increase spending.** Members and normal customers are split almost 50/50 (501 vs 499), and their average baskets are $327.8 and $318.1 (p = 0.59, not significant). Ratings are also the same. Membership is not currently changing behaviour.

14. **Payment methods are balanced, with cash slightly ahead.** Cash brings 34.7% of revenue, Ewallet 34.1% and Credit card 31.2%. Average baskets are within $7 of each other, so payment method does not signal who spends more.

15. **High-value outliers are genuine and spread across the stores.** 9 baskets exceed the IQR upper fence (max $1,042.65). Every one of them is 10 items at a unit price of about $95-99, so they are real maximum-size purchases and not data errors. They appear in all three branches and in five of the six categories, so these buyers are worth treating as VIPs.

16. **The profit columns carry no information.** `gross income` equals the 5% tax amount and the gross margin is a constant 4.7619% for every row. Because of this the dataset cannot show which category or branch is actually more profitable, which is a limitation of the data itself.

11. Business Recommendations

1. **Make the membership programme earn its cost (Insight 13).** Members spend no more than normal customers, so the card is not creating value yet. I would add spend-based tiers or points that reward a larger basket (for example a bonus above $400, a level only about the top 20% of baskets reach), and track the member vs non-member basket gap every month.

2. **Staff and promote around the time-of-day curve (Insights 10 and 11).** Put the strongest staffing and fresh stock at the 19:00 peak and on Saturdays. Run time-limited offers in the quiet slots (16:00-17:00, 20:00) and on Mondays, since those are the hours with the most unused capacity.

3. **Run a Health and beauty pilot (Insights 7, 8 and 9).** It is the weakest category overall, and the gap between Branch B ($20.0k) and Branch A ($12.6k) suggests there is room to grow. Because the branch and gender differences are not statistically proven, I would test before investing: copy Branch B's range and display in Branch A for a month, run a women-focused promotion in one branch only, and compare results against the other branches.

4. **Test a Home and lifestyle fix in Branch C (Insight 9).** C earns $13.9k there compared with $22.4k in A. Review what A stocks and how it displays it, trial the same layout in C, and keep the change only if the numbers improve.

5. **Increase items per basket with bundles and multi-buy offers (Insight 3).** Quantity is one of the two independent drivers of basket value, and Fashion accessories has the lowest average quantity (5.07) with the highest unit price. "Buy 3, save 10%" style offers in that category give the largest headroom.

6. **Investigate Branch B's satisfaction and collect better feedback (Insights 4, 5 and 6).** Rating does not depend on anything we measure, so more spend or more traffic will not fix it. Start a short post-purchase survey (staff, wait time, product availability) at Branch B first, and use it to find the real drivers.

7. **Record real cost and margin per product (Insight 16).** Pricing strategy cannot be optimised when the margin is the same constant for all rows. Capture true cost by product line, then repeat this analysis on actual profit.

12. Code and Output Evidence

These blocks show the actual code from my notebook and the output it produced. The complete notebook, with every cell, is in the GitHub repository.

Importing libraries and loading the dataset

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
import os, warnings
warnings.filterwarnings('ignore')

sns.set_theme(style='whitegrid', palette='deep')
plt.rcParams['figure.dpi'] = 100
os.makedirs('graphs', exist_ok=True)

def save(name):
    # one helper so every graph is saved as an image for the report
    plt.tight_layout()
    plt.savefig(f'graphs/{name}.png', dpi=130, bbox_inches='tight')
    plt.show()

df = pd.read_csv('data/supermarket_sales.csv')
print('Loaded OK')
```

```
Loaded OK
```

Dataset shape

```
print('Shape (rows, columns):', df.shape)
```

```
Shape (rows, columns): (1000, 17)
```

Basic statistics

```
df.describe().T.round(3)
```

```
                         count     mean      std     min      25%      50%  \
Unit price              1000.0   55.672   26.495  10.080   32.875   55.230
Quantity                1000.0    5.510    2.923   1.000    3.000    5.000
Tax 5%                  1000.0   15.379   11.709   0.508    5.925   12.088
Total                   1000.0  322.967  245.885  10.678  124.422  253.848
cogs                    1000.0  307.587  234.177  10.170  118.498  241.760
gross margin percentage 1000.0    4.762    0.000   4.762    4.762    4.762
gross income            1000.0   15.379   11.709   0.508    5.925   12.088
Rating                  1000.0    6.973    1.719   4.000    5.500    7.000

                            75%       max
Unit price                77.935    99.960
Quantity                   8.000    10.000
Tax 5%                    22.445    49.650
... (output shortened)
```

Missing value check

```
print('Missing values per column:')
print(df.isnull().sum())
print('\nTotal missing:', df.isnull().sum().sum())
```

```
Missing values per column:
Invoice ID                 0
Branch                     0
City                       0
Customer type              0
Gender                     0
Product line               0
Unit price                 0
Quantity                   0
Tax 5%                     0
Total                      0
Date                       0
Time                       0
Payment                    0
cogs                       0
gross margin percentage    0
gross income               0
Rating                     0
dtype: int64

Total missing: 0
```

Verification after cleaning

```
# Verify the dataset after cleaning
print('Shape:', df.shape)
print('Missing:', df.isnull().sum().sum(), '| Duplicates:', df.duplicated().sum())
print('Date range:', df['Date'].min().date(), 'to', df['Date'].max().date(), '(', df['Date'].nunique(), 'trading days
```

```
Shape: (1000, 21)
Missing: 0 | Duplicates: 0
Date range: 2019-01-01 to 2019-03-30 ( 89 trading days )
<class 'pandas.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 21 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   Invoice ID               1000 non-null   str
 1   Branch                   1000 non-null   str
 2   City                     1000 non-null   str
 3   Customer type            1000 non-null   str
 4   Gender                   1000 non-null   str
 5   Product line             1000 non-null   str
 6   Unit price               1000 non-null   float64
 7   Quantity                 1000 non-null   int64
 8   Tax 5%                   1000 non-null   float64
 9   Total                    1000 non-null   float64
 10  Date                     1000 non-null   datetime64[us]
 11  Time                     1000 non-null   str
 12  Payment                  1000 non-null   str
 13  cogs                     1000 non-null   float64
 14  gross margin percentage  1000 non-null   float64
 15  gross income             1000 non-null   float64
 16  Rating                   1000 non-null   float64
 17  Hour                     1000 non-null   int32
 18  Month                    1000 non-null   str
 19  Weekday                  1000 non-null   str
 20  Week                     1000 non-null   datetime64[us]
dtypes: datetime64[us](2), float64(7), int32(1), int64(1), str(10)
... (output shortened)
```

Correlation matrix

```
corr_cols = ['Unit price', 'Quantity', 'Total', 'Rating', 'Hour']
corr = df[corr_cols].corr().round(3)
corr
```

```
            Unit price  Quantity  Total  Rating   Hour
Unit price       1.000     0.011  0.634  -0.009  0.008
Quantity         0.011     1.000  0.706  -0.016 -0.007
Total            0.634     0.706  1.000  -0.036 -0.003
Rating          -0.009    -0.016 -0.036   1.000 -0.031
Hour             0.008    -0.007 -0.003  -0.031  1.000
```

13. Challenges Faced

- **Columns that look like data but are formulas.** I only noticed that gross income equals the tax column and that the margin is constant after checking the calculations. I had to exclude them from correlation analysis and accept that profitability cannot be studied with this dataset.

- **Differences that look large but are not significant.** Several charts, for example revenue by branch and category or by gender, show clear gaps. When I tested them, they were within normal random variation for 1,000 rows. I had to word my insights carefully and present those findings as ideas to test rather than facts.

- **Avoiding graphs just for the count.** The task needs 12 or more graphs, but each one has to say something. I chose charts that answer a specific question (time of day, weekday, branch against category) and wrote what each one shows.

- **Partial time periods.** The first and last weeks and a shorter February made the weekly and monthly charts look misleading. I compared revenue per trading day to get a fair monthly comparison.

- **No real profit or cost data.** This limits the pricing recommendations, so one of my recommendations is to record real product costs.

14. Conclusion

The analysis shows a well-balanced supermarket business: the branches, product lines, payment methods and customer types all perform at similar levels. The clearest patterns are that basket size comes from quantity and price working independently, that there are strong time-of-day and weekday cycles (peak at 19:00 and on Saturdays), and that membership does not increase spending. Satisfaction ratings are not explained by anything in the data, which suggests that service factors that were not recorded matter more. Differences between genders and branches in category spending exist in the sample, but need more data to confirm. The main limitations are that the data covers only three months and has no real profit information. A natural next step is feature engineering and predictive modelling for customer rating or daily sales.

15. GitHub Repository Link

[paste your repository link here]